Conductor-changing prime-degree isogenies of ECC2K-130

Report dated 2026-10-07, extended 2026-10-08 (§8). Study directory `research/ecc2k130_conductor_isogenies_20261007/`. Standalone study: no `H-*`, `EV-*`, `DEC-*` or `RUN-*` record was written. Run artifacts are the `.out` files in the directory. Labels: **DERIVED** proof or exact computation from stated facts; **MEASURED** observed in this study's runs; **INDEPENDENTLY VERIFIED** confirmed by a second, independent method; **PROPOSED** a hypothesis, not shown here.

1. Question and scope

Which prime-degree isogenies from ECC2K-130 change the conductor of the endomorphism ring, and which of them can be computed? The scope is the whole $a_2 = 0$ isogeny class over $\mathbb{F}_{2^{131}}$, all prime degrees.

2. Object

- $E : y^2 + xy = x^3 + 1$ over $\mathbb{F}_2[x]/(x^{131} + x^8 + x^3 + x^2 + 1)$. A curve E_b is named by the integer of the bits of b. **DERIVED**
- $\#E = 4\ell$ with $\ell = 68056473384187692693232012949340998512 9$ prime; the trace is -22283658519494248867. **DERIVED**
- $\mathrm{End}(E) = \mathcal{O}_K$ with $K = \mathbb{Q}\left(\sqrt{-7}\right)$ and $h = 1$. The Frobenius conductor is $f_\pi = [\mathcal{O}_K : \mathbb{Z}[\pi]] = 263 \cdot 146505763881528721$; the prime 263 splits and the other is inert. **DERIVED**

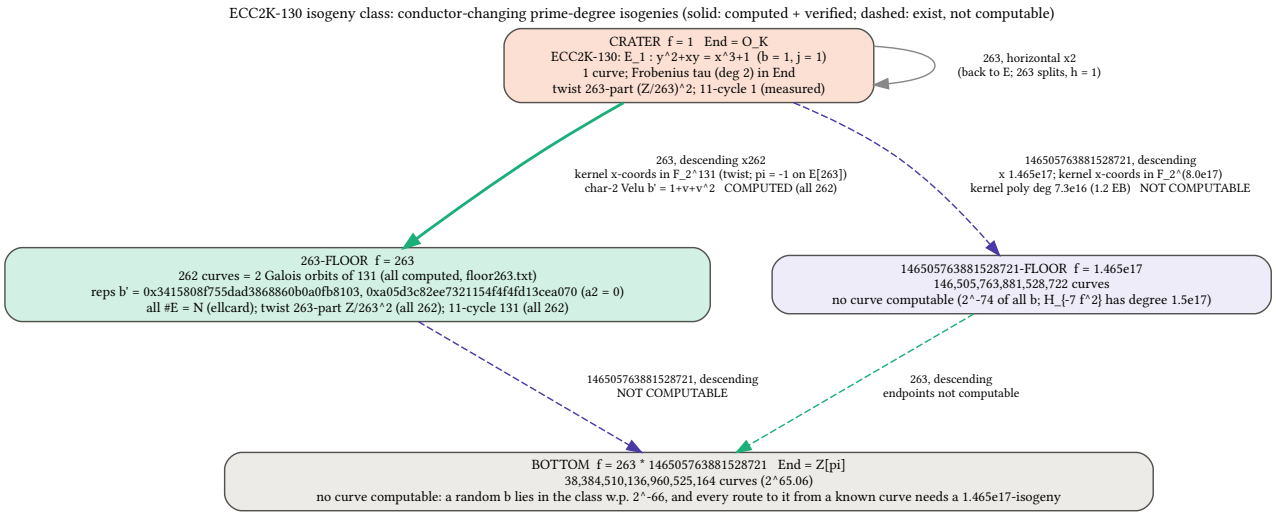


Figure 1: The isogeny class by level. Solid: computed here, with endpoints verified. Dashed: exists by theory but cannot be computed. Source: `volcano_ecc2k130.dot`.

3. Which prime degrees can change the conductor

An ℓ-isogeny between ordinary curves changes the conductor by a factor of at most ℓ, and can change it only if $\ell \mid f_\pi$ (Kohel). Every other prime degree gives a horizontal isogeny, with conductor ratio 1. Here that leaves exactly two degrees: **263** and **146505763881528721**. **DERIVED**

4. The 263-isogenies (computed)

- $\pi \equiv -1 \pmod{263}$, so $E[263]$ is $\mathbb{F}_q$-rational on the quadratic twist $E_t : y^2 + xy = x^3 + x^2 + 1$, with the same x-coordinates. No extension field is needed. **DERIVED**
- $\#E_t = 2 \cdot 263^2 \cdot 196783164088501186057678526575102 39$ and $E_t(\mathbb{F}_q) \cong \mathbb{Z}/(\#E_t/263) \times \mathbb{Z}/263$, so $E_t[263]$ is all rational. **MEASURED**
- All 264 kernels, via char-2 Vélu ($b' = 1 + v + v^2$, checked 6/6 against PARI `ellisogeny` at $m = 37$):
  - 2 map back to E. These are the horizontal isogenies; since 263 splits and $h = 1$, they are endomorphisms. **MEASURED**
  - 262 give 262 distinct curves, the whole 263-floor ($h(\mathcal{O}_{263}) = 262$). Every one has $\#E = N$ by `ellcard`. **MEASURED**

- **Level checks.** Each image's twist has a cyclic $\mathbb{Z}/263^2$ 263-part (262/262), against $(\mathbb{Z}/263)^2$ on the crater. The horizontal 11-isogeny cycle has length 131 (262/262), equal to $\mathrm{ord}[\mathfrak{l}_{11}]$ in $\mathrm{Cl}(-7 \cdot 263^2)$; on the crater it is 1. **INDEPENDENTLY VERIFIED**
- The 262 curves form 2 Galois orbits of 131, with representatives $b' =$ `0x3415808f755dad3868860b0a0fb8103` and `0xa05d3c82ee7321154f4f4fd13cea070` ($a_2 = 0$). **MEASURED**

5. The 146505763881528721-isogenies: why they cannot be computed

All $\ell + 1 = 146505763881528722$ of them exist and descend. **DERIVED** Every route to one is out of reach:

Route	Barrier
Kernel points (Vélu, √élu)	$\pi \equiv c \pmod{\ell}$, with $\mathrm{ord}(c) = 1.22 \cdot 10^{16}$ and $-1 \in \langle c \rangle$. The kernel x-coordinates first appear in $\mathbb{F}_{2^{131k}}$ with $k = 6104406828397030$, where one element takes $8.0 \cdot 10^{17} \approx 2^{59.5}$ bits. **DERIVED**
Kernel polynomial over $\mathbb{F}_{2^{131}}$	Degree $(\ell - 1)/2 = 7.3 \cdot 10^{16}$; about 1.2 EB just to store. **DERIVED**
Modular polynomial $\Phi_\ell(1, Y)$	Degree $\ell + 1 \approx 1.5 \cdot 10^{17}$, with integer coefficients of order $\ell \log \ell$ bits. **DERIVED**
CM / Hilbert class polynomial	Discriminant $-7\ell^2$, degree $h = 1.5 \cdot 10^{17}$. **DERIVED**
Random search for an image	A random b lands on that floor with probability 2^{-74}. **DERIVED**
√élu	About $\sqrt{\ell} \approx 2^{28.5}$ operations, but each is in the $2^{59.5}$-bit field. **DERIVED**

Every route costs at least about 2^{56}. Pollard rho on ECC2K-130 with Frobenius costs about $2^{60.8}$, so just writing one of these isogenies down costs about as much as solving the discrete logarithm. The same barrier holds for the ℓ-isogenies from the 263-floor and for every curve at the bottom: a random b lies in the class at all with probability 2^{-66}, and every route from a known curve passes through an ℓ-isogeny. **DERIVED**

6. Analogues: m = 83, m = 109, NIST K-163 (`analogues/`)

For each conductor prime ℓ', $\pi \equiv c$ on $E[\ell']$, and the kernel lives in $\mathbb{F}_{2^{mk}}$, where k is the least integer with $c^k = \pm 1$. **DERIVED**

Curve	Prime, k	Result
ECC2K-130 ($m = 131$)	263, $k = 1$	All 262 computed (this report). **MEASURED**
$m = 83$, $a = 0$	6473, $k = 3236$	Kernel field $\mathbb{F}_{2^{268588}}$; PARI cannot build it in 4 GB **MEASURED** about 7 weeks extrapolated **PROPOSED**. **Not computed.**
$m = 109$, $a = 1$	3271, $k = 5$	6 computed in under 1 s each, 6 distinct orbits, $\#E' = N$; 23-cycle 545. **MEASURED** **INDEPENDENTLY VERIFIED**
NIST K-163	45641, $k = 20$	6 computed, about 1 min each; $\#E' = N$; 11-cycle 9128. **MEASURED** **INDEPENDENTLY VERIFIED**
NIST K-163	82153, $k = 63$	6 computed, about 15 min each; $\#E' = N$; 11-cycle 10269. **MEASURED** **INDEPENDENTLY VERIFIED**

ECC2K-130's 263 is unusually cheap: $k = 1$. $m = 83$ has no such prime, while K-163 has two computable ones. **DERIVED**

7. ECDLP consequence (scoped) and limitations

- The 263-isogenies make transferring a discrete logarithm to the 263-floor essentially free. **DERIVED**
- A 263-floor curve has no Frobenius endomorphism; its cheapest non-integer endomorphism has degree $\geq 7 \cdot 263^2/4 \approx 121045$. **DERIVED**
- The volcano studies at $m = 37$ and $m = 41$ measured that rho on such curves runs with negation only, $\approx \sqrt{m}$ times more steps than the crater with Frobenius. At $m = 131$ that predicts about $11.4\times$ more steps. That is an extrapolation, not measured here. **PROPOSED**

8. Extensions (2026-10-08): the CM route, $m = 51/53$, a second level on K-163

Details and all run outputs are in `EXTENSIONS.md` and in the directories `cm_route/`, `survey/`, `m51_m53/` and `second_level/`.

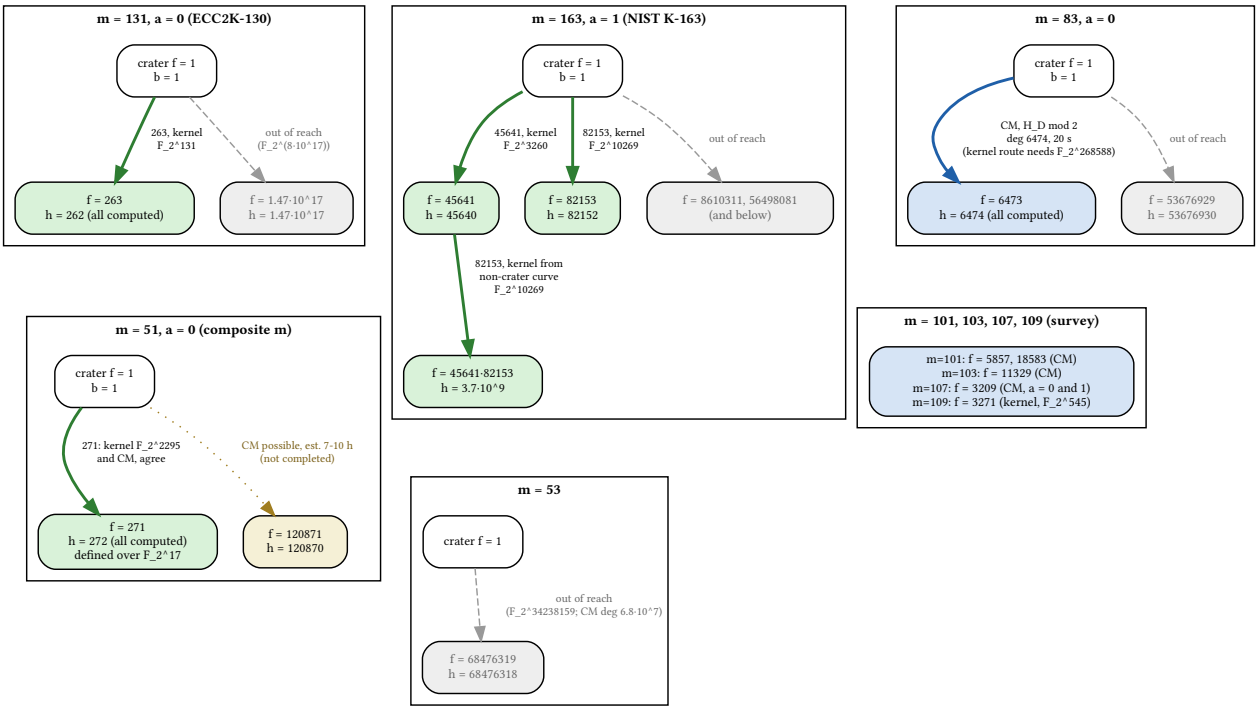


Figure 2: Conductor levels reached on each Koblitz class, and the route used. Green: kernel route (Vélu on kernel points). Blue: CM route. Dotted: estimated, not completed. Grey dashed: out of reach. Source: `reachability.dot`.

CM route. The F-floor of a Koblitz crater consists of the roots mod 2 of the class polynomial for $D = -7F^2$. PARI `polclass(D, 1)` uses the Weber f invariant (allowed because $D \equiv 1 \pmod 8$ and $3 \nmid D$). Mod 2 the relation $\left(f^{24} - 16\right)^3 = jf^{24}$ becomes $j = f^{48}$, so each root r gives $b = r^{-48}$. **DERIVED** No kernel field is needed.

An earlier draft read the roots as γ_2 and used $b = r^{-3}$. That labelling was wrong, but the sets were still exact: $r^{-3} = \left(r^{-48}\right)^{1/16}$ is a Galois conjugate, and every floor is Galois-stable. The corrected scripts reproduce the same files byte for byte. **MEASURED**

- **Validation.** At $m = 37$ ($f = 73$) and $m = 41$ ($f = 409$), the root sets equal the floors enumerated by explicit isogenies: 74/74 and 410/410. **INDEPENDENTLY VERIFIED**
- **$m = 83$, the 6473-floor.** The kernel route needs $\mathbb{F}_{2^{268588}}$. By the CM route, `polclass` takes 20 s and gives 6474 roots in $\mathbb{F}_{2^{83}}$, all with $\#E = N$, in 78 Galois orbits of 83. **MEASURED** For the level: $\mathrm{ord}[\mathfrak{l}_{11}] = 1079$ in $\mathrm{Cl}(-7 \cdot 6473^2)$, and six 11-cycles of length 1079 tile all 6474 curves. **INDEPENDENTLY VERIFIED**
- **Survey.** The floors $m = 101$: 5857 and 18583; $m = 103$: 11329; $m = 107$: 3209 (for $a = 0$ and $a = 1$) were all computed in 6–313 s each. Every root has a model with $\#E = N$, and the sampled 11-cycles equal $\mathrm{ord}[\mathfrak{l}_{11}]$. **INDEPENDENTLY VERIFIED**
- **Cost.** `polclass` time grows as about $|D|^{1.2-1.3}$ (3 sizes, one run each), so $h \approx 10^5$ is the practical limit here. **MEASURED**

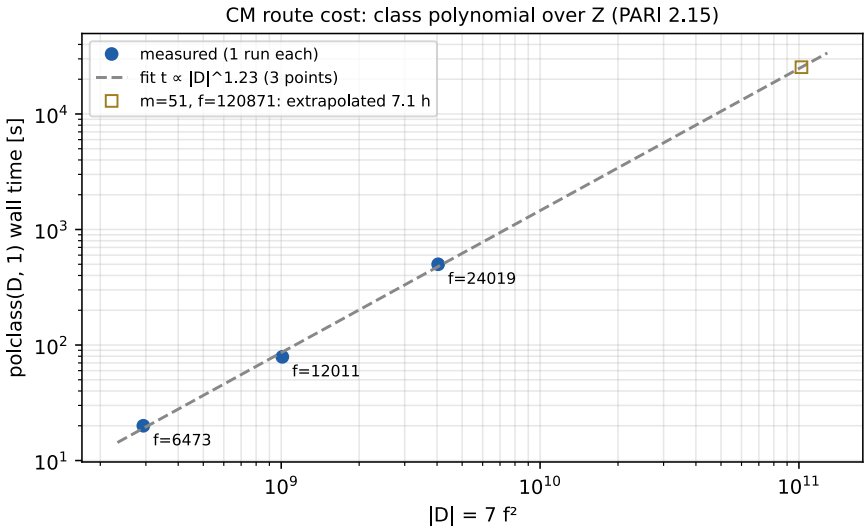


Figure 3: `polclass(D, 1)` wall time against $|D|$; one run per point; dashed: power-law fit; square: extrapolation to the $m = 51$ 120871-floor. Source: `cm_route/scale_polclass.out`, `cm_route/plot_scaling.py`.

$m = 51$ and $m = 53$.

- At $m = 51$, $f_\pi = 271 \cdot 120871$.
- **The 271-floor.** 271 is the conductor of $\mathbb{Z}[\tau^{17}]$, so the floor is defined over $\mathbb{F}_{2^{17}}$ (orbits of 17). **DERIVED** Both routes compute it. CM gives 272 curves; 40 explicit 271-isogenies with kernels in $\mathbb{F}_{2^{2295}}$ all land in that set (40/40), reaching 14 of the 16 orbits. **INDEPENDENTLY VERIFIED**
- **The 120871-floor.** The kernel route needs $\mathbb{F}_{2^{60435}}$. The CM route needs $h = 120870$, estimated at 7–10 h. It was started twice, and container restarts killed it both times; no result is claimed. **PROPOSED**
- **$m = 53$.** $f_\pi = 68476319$: the kernel route needs $\mathbb{F}_{2^{34238159}}$ and the CM route a degree-$6.8 \cdot 10^7$ class polynomial. Out of reach. **DERIVED**

Second level on K-163. `vert_down.gp` descends from a non-crater curve. The curve's b is embedded in the kernel field through a root of the field modulus in the subfield generated by $w = z^{(2^{mk}-1)/(2^m-1)}$.

- **Validation at $m = 41$.** From a 409-floor curve, two 1721-isogenies land on curves with 23-cycle 2870 and 11-cycle 1722, the bottom-level values. **INDEPENDENTLY VERIFIED**
- **K-163.** From the 45641-floor curve $b' =$ `2018301825331905268970411624754627525843172028 5`, one 82153-isogeny (kernel in $\mathbb{F}_{2^{10269}}$) gives $b'' =$ `1159925961055100583252080443667523394222805212 27` ($a_2 = 1$, $\#E = N$). **MEASURED**
- **Level of b''.** Its 45641-part and 82153-part are both cyclic ($\mathbb{Z}/\ell'^2$), against rank 2 for both on the crater and rank 2 for the 82153-part on b'. So both primes divide the conductor of its endomorphism ring, and the conductor is $45641 \cdot 82153$. Its horizontal 11-cycle has length 82152, which is ord$[\mathfrak{l}_{11}]$ at that level (crater 1, 45641-floor 9128, 82153-floor 10269). **INDEPENDENTLY VERIFIED** A rerun with PARI's default seed reproduced b'' exactly. A run with a different seed gave a second, independent image, $b''' =$ `52997549064096697226797992255204505576485896783`, in another Galois orbit; it passes the same two checks (both torsion parts cyclic; 11-cycle 82152). **INDEPENDENTLY VERIFIED**
- **Four more.** Four further descents, each from a different 45641-floor curve with its own seed, gave four more level-2 curves (`second_level/down163_s1..4.txt`). Each passes both checks: 11-cycle 82152, and both torsion parts cyclic. All six level-2 curves lie in distinct Galois orbits. **INDEPENDENTLY VERIFIED**
- **Batch of twenty.** Twenty more descents (`second_level/batch20/`), rotating over the six 45641-floor representatives with seeds 101–120, gave twenty more level-2 curves (839–1323 s each). All twenty pass both checks (11-cycle 82152; both torsion parts cyclic), and none shares a Galois orbit with another run or with the earlier six. That makes 26 level-2 curves in distinct orbits, out of 23,002,560. **INDEPENDENTLY VERIFIED**

Limits. The CM route gives the image curves, not the isogenies themselves; no 6473-isogeny of $m = 83$ was written down. **PROPOSED**

9. Next checks

- Run the `audit-curve` skill on one representative from each 263-floor orbit: twist, MOV/FR, anomalous and GHS descent checks. **PROPOSED**